Civics Group	Index Number	Name (use BLOCK LETTERS)

H2

ST. ANDREW'S JUNIOR COLLEGE
2021 JC2 PRELIMINARY EXAMINATIONS

H2 BIOLOGY **9744/2**

Paper 2 Mark Scheme

Monday 30th August 2021 2 hours

Materials: Question Paper

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.
Write your answers in the spaces provided on the question paper.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiners' Use	
1	/10
2	/12
3	/10
4	/9
5	/6
6	/11
7	/12
8	/12
9	/11
10	/7
Total	**/100**

Conceptual error (C)	Data Quoting (D)	Expression (E)	Misreading the question (Q)

This document consists of **xx** printed pages and **x** blank page.

[Turn over

Question 1

(a) T-helper lymphocytes and Leydig cells are two types of mammalian cells. The main role of T-helper lymphocytes and Leydig cells is to synthesise and secrete cell-signalling molecules.

- T-helper lymphocytes synthesise proteins known as cytokines.
- Leydig cells synthesise the steroid (lipid) hormone testosterone from cholesterol. Leydig cells also synthesise cholesterol.

Figure 1.1 shows part of a mammalian cell.

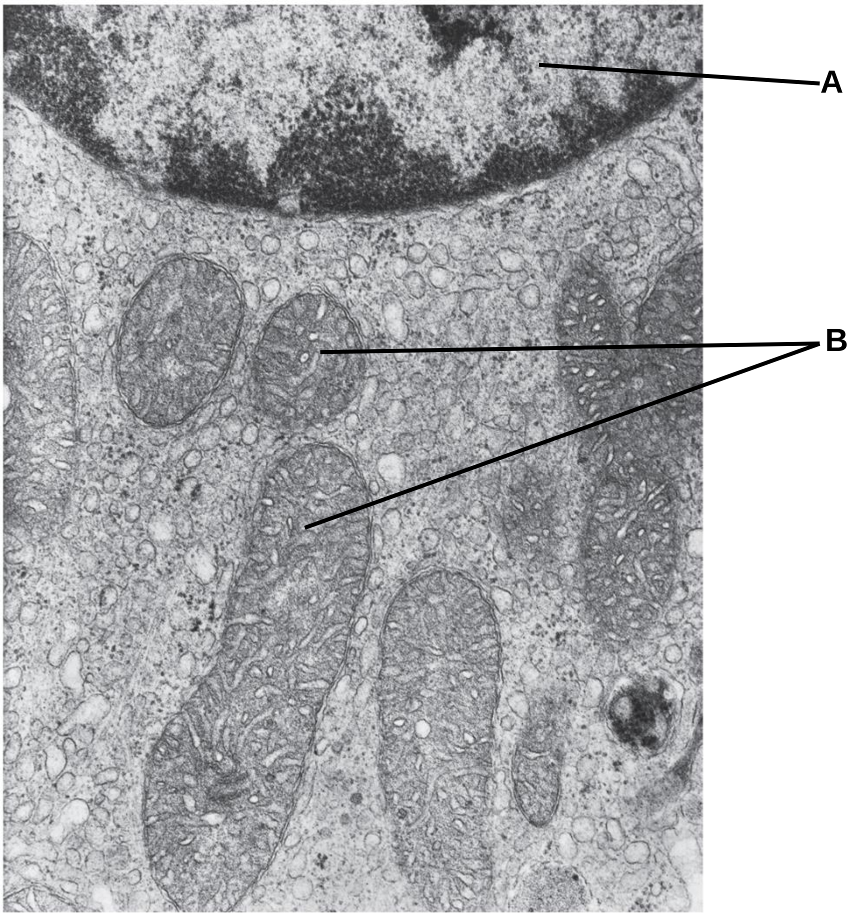


Fig. 1.1

(i) Identify organelles **A** and **B** on Fig. 1.1. [1]

A: nucleus
B: mitochondr**ia**

(ii) Organelle **A** is involved in mitotic cell cycle. State the process that occurs during interphase to produce genetically identical daughter cells.

……………………………………………………………………………………[1]

1 Semi-conservative DNA replication ;

(iii) Explain whether Fig. 1.1 shows part of a Leydig cell or part of a T-helper lymphocyte.

……………………………………………………………………………………...[2]

1. *[State]* Leydig cell ;
2. *[Explanation]* much / many / high proportion of, smooth endoplasmic reticulum ; (sER present) that functions to **synthesise**, cholesterol / testosterone / steroid hormones / **lipids** ;

   OR

   no / little, rough endoplasmic reticulum ; (rER present) that function to **synthesise polypeptide** at ribosomes on its surface / **chemical modification** of protein in the lumen of rER ;

(b) Cell-surface receptors are membrane-anchored (integral) proteins that bind to external signalling molecules.

Such receptors are synthesised within the cell and brought to the cell surface membrane as shown in the Figure 1.2.

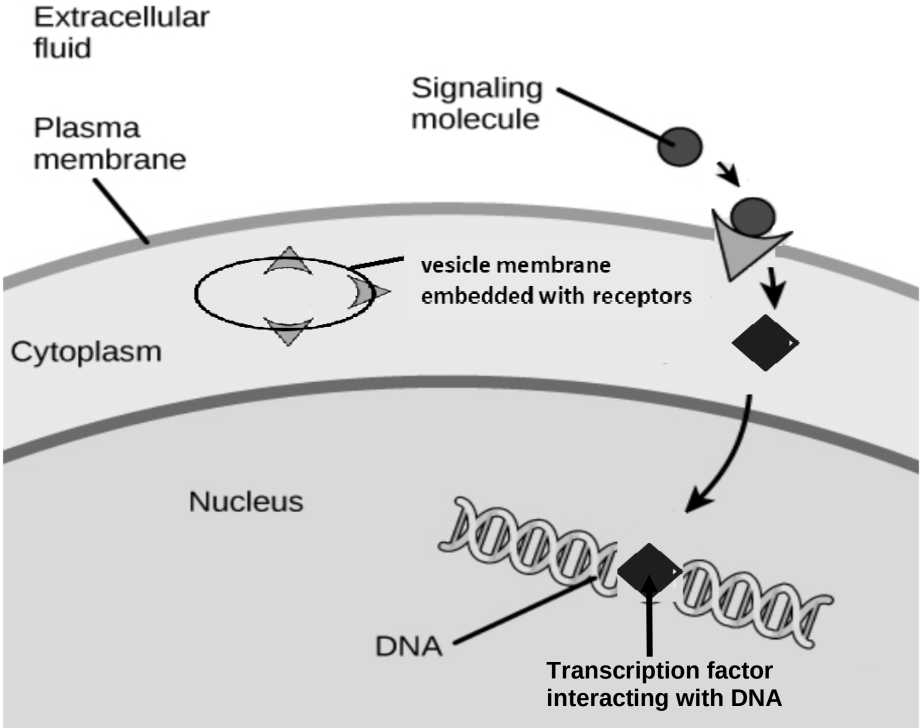


Fig. 1.2

With reference to Fig. 1.2, describe how the receptor is brought to and become part of the cell surface membrane.

…………………………………………………………………………………..................[2]

1. Secretory vesicles **with embedded receptors** (with ligand-binding site of the receptor projecting into the vesicle) **move towards** the cell surface membrane (of cells);
2. Vesicle's membrane fuses with the **cell surface** membrane ; resulting in the **ligand-binding site** of the receptor **projecting into the extracellular fluid** *[visually obtained from Fig. 1.1]*
3. Ref. ATP expenditure for **movement** of vesicles on the **microtubules / filaments of the cytoskeleton** ;

(c) *Eichhornia crassipes*, a water hyacinth plant, is an invasive species found in lakes and rivers throughout Asia. They grow rapidly, producing many flowers and seeds. Each flower of this plant has six anthers (each anther is at one end of the filament) and one stigma as shown in Figure 1.3.

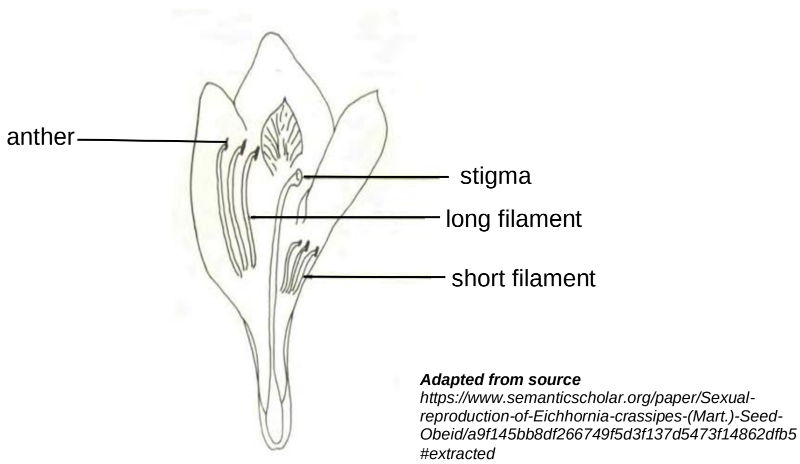


Fig. 1.3 showing spatial arrangement of stamen (comprising of both anther and filament) and stigma in *E. crassipes*.

Each anther was observed to release many pollen grains. After pollination, the pollen grains germinate after it comes into contact with the stigma, in response to the sugary fluid secreted by the mature stigma. A pollen tube grows out from each pollen grain. The pollen tube serves to transport the male gamete cells to the female gamete cells (ovules) in the flower.

The tip of the pollen tube is protected by an outer layer of pectin.

Pectin is a polysaccharide.

Starch grains, mitochondria and vesicles are present in the tip of the pollen tube.

Using the information given and your content knowledge on organelles and carbohydrates, describe how the production of pectin occurred at the tip of a pollen tube.

…………………………………………………………………………………………[4]

1. Starch is {digested / broken down / hydrolysed} to release glucose ;
2. **Condensation** reactions with **release** of 1 water molecule per glycosidic bonds are used to form pectin ;
3. Pectin {secreted into / enclosed in / transport in} vesicles ;
4. Vesicles **fuse** with the {tip / cell surface membrane} to **release** {pectin / content} ; [**ACCEPT**: reference to exocytosis releasing contents]
5. Cellular respiration in mitochondria **releases ATP** for production of {vesicles / pectin / enzymes / other relevant molecules} ;
   [**ACCEPT**: energy used for movement of vesicles in exocytosis]

[Total: 10]

QUESTION 2

Rhodopsin is a cell signalling receptor found in the rod cells of the retina that is associated with retinal thinning and degeneration. There have been correlations established between retinal thinning and Alzheimer's disease. Hence, scientists have realised the potential of the study of rhodopsin in the early detection and progression monitoring of Alzheimer's disease.

Figure 2.1 shows the signalling pathway of rhodopsin in rod bipolar cells.

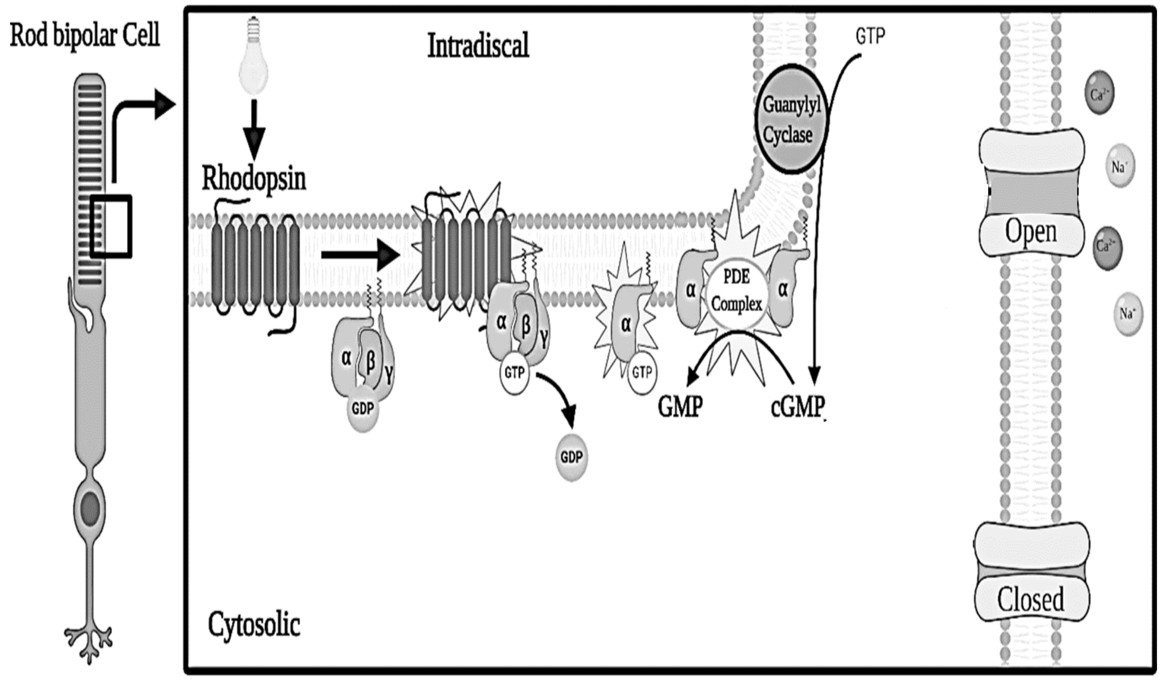


Fig. 2.1

(a) (i) With reference to Fig. 2.1, state the class of receptors in which rhodopsin belongs to. Justify your answer.

....[2]

1. G protein {linked/coupled} receptor ;

2. the receptor has 7 **transmembrane** {helices / domains}

(ii) Explain how the structure of rhodopsin relates to its function.

………………………………………………………………………………………..…………..[4]

Any 2 pairs:

1. [Structure] {**Extracellular domain** / ligand binding site} is **complementary in shape** to a specific ligand.
2. [Function] **Allows** chemical **message from extracellular environment to be received by target cells** when ligand binds receptor.

3. [Structure] {**Intracellular domain** (which is exposed upon receptor activation and) / G protein binding site} is **complementary in shape** to a G-protein
4. [Function] Allows the binding of G-proteins to the G-protein binding site, which causes **G-protein activation**, and subsequently results in the **relay of signals** from the receptors at the cell surface membrane to target molecules within the cell.
5. [Structure] GPCR is a transmembrane protein containing amino acids with non-polar R groups that interact with the **non-polar hydrophobic core of the membrane**; while the polar and charged R groups interact with the **hydrophilic polar phosphate heads** of the membrane.
6. [Function] Allows GPCR to **function on both sides of the membrane** i.e. it can bind signal molecule in the extracellular environment and generates intracellular signals on the cytosolic environment.

When light activates rhodopsin, it initiates a sequence of events that consequently decreases the concentration of cGMP. When the cGMP concentration decreases, the cGMP-gated channels will close, preventing depolarization of the membrane induced by the influx of Na^{+} and Ca^{2+}.

(b) With reference to Fig. 2.1, describe how activation of rhodopsin can lead to a decrease in cGMP concentration.

……………………………………………………………………………………..…………..[5]

1. When light strikes rhodopsin, it triggers a **conformational change** in the G-protein linked receptor ;
2. **exposing** the **cytosolic** G-protein binding domain of the G-protein linked receptor, (hence activating the receptor) ;
3. G-protein binds to G-protein binding domain and **exchanges** GDP for GTP ; GTP-bound G-protein is **activated**
4. α subunit attached to GTP **dissociates** from the receptor and moves along the cell surface membrane ;
5. (α subunit) **binds** to PDE complex and **activates** it.
6. GTP is converted to cGMP with the help of guanylyl cyclase ;
7. PDE complex **hydrolyses** cGMP to GMP, (resulting in the decrease in concentration of cGMP).

[Max 5]

Over-activation of rhodopsin from light results in prolonged depolarization in membrane potential. This could accelerate retinal thinning and degeneration. As such, over the years, cells have evolved to develop a mechanism to prevent the over-activation of rhodopsin.

(c) Suggest one mechanism that the cell can use to prevent the over-activation of rhodopsin.

……………………………………………………………………………………..…………..[1]

1. Endocytosis / **Invagination of the membrane** containing rhodopsin to prevent over-stimulation.

   [Reject: ideas related to mutations resulting in change in 3D conformation of rhodopsin cos usually cells do not mutate proteins in order to switch off its activity]

   AVP: compound (naturally **synthesized** and **released** by cells) that can bind to receptor via complementary 3D conformation to prevent activation of receptor

[Total: 12]

QUESTION 3

ATP is often known as the universal ‘energy currency’ of the cell.

Most ATP are made in cells by membrane systems that create proton gradients by pumping protons from one compartment to another.

Figure 3.1 shows two of such membrane systems.

Legend:
represents ATP synthase

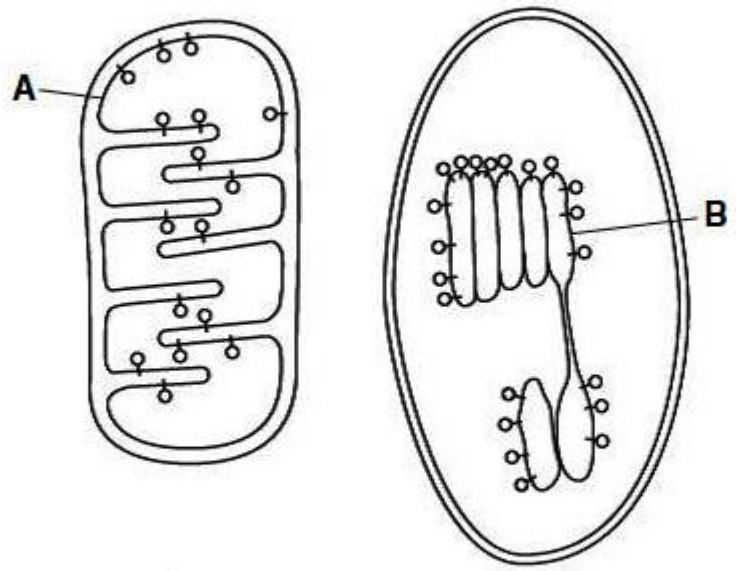
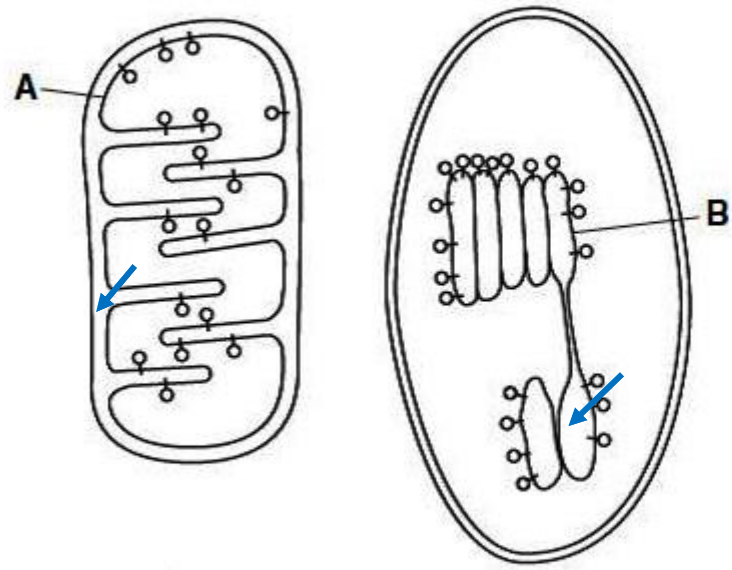


Fig. 3.1

(a) (i) Draw arrows onto **each** of the membrane systems in Fig. 3.1 to show the direction in which protons are pumped.
....[1]

1. A – from matrix into intermembrane space, B – from stroma into thylakoid space; [Reject: if arrow goes through ATP synthase]



(ii) Outline how energy is generated for the pumping of protons across membrane system B.
………[3]

1. [ETC] Electrons from **special chlorophyll a pigments** [reject: NADP] are passed down electron carriers;
2. electron carriers are arranged in progressively **decreasing energy levels**;
3. **Energy released** by **transfer of electrons / electron flow** is used to pump H+ from stroma to thylakoid space.

(iii) Explain why transmembrane proteins are necessary for protons to be pumped across the membrane systems.
……………………………………………………………………………………………………[2]

1. [Chem. property of protons] Protons are charged ions [ignore: polar] ;
2. They are therefore **unable to pass through** the **hydrophobic core** of the membranes ;

It is also known that there are cyclic pathways involving the use of enzymes occurring in both membrane systems.

(b) State the names of the two cyclic pathways and outline the differences between them.
....[4]

1 State pathways	Calvin cycle	Krebs cycle
Differences		
2 Location	Stroma; of chloroplast	Mitochondrial matrix; of mitochondria
3 Electron/hydrogen carriers involved	$NADP^+$ / NADPH	NAD^+/NADH, FAD/$FADH_2$
4 Carbon dioxide used/released	**CO_2** enters the Calvin cycle, and is **fixed** by **combining** with a 5C compound called **ribulose bisphosphate (RuBP)**, catalysed by the enzyme rubisco (RuBP carboxylase)	**CO_2** is **released** during **oxidative decarboxylation**
5 ATP used/synthesized	**Used** in reduction phase when **PGA is reduced to PGAL**; and used in **regeneration of RuBP** from PGAL	**Synthesised** by **substrate level phosphorylation** [Reject: oxidative phosphorylation]
6 Regenerated product	RuBP	Oxaloacetate

[Total: 10]

Question 4

(a) Figure 4.1 shows the nematode worm, *Caenorhabditis elegans*.

magnification ×250

Fig. 4.1

Stem cells of *C. elegans* have been studied.

Figure 4.2 shows the change in mass of DNA per nucleus in a stem cell during one cell cycle. 1pg = 1 × 10^{-12}g.

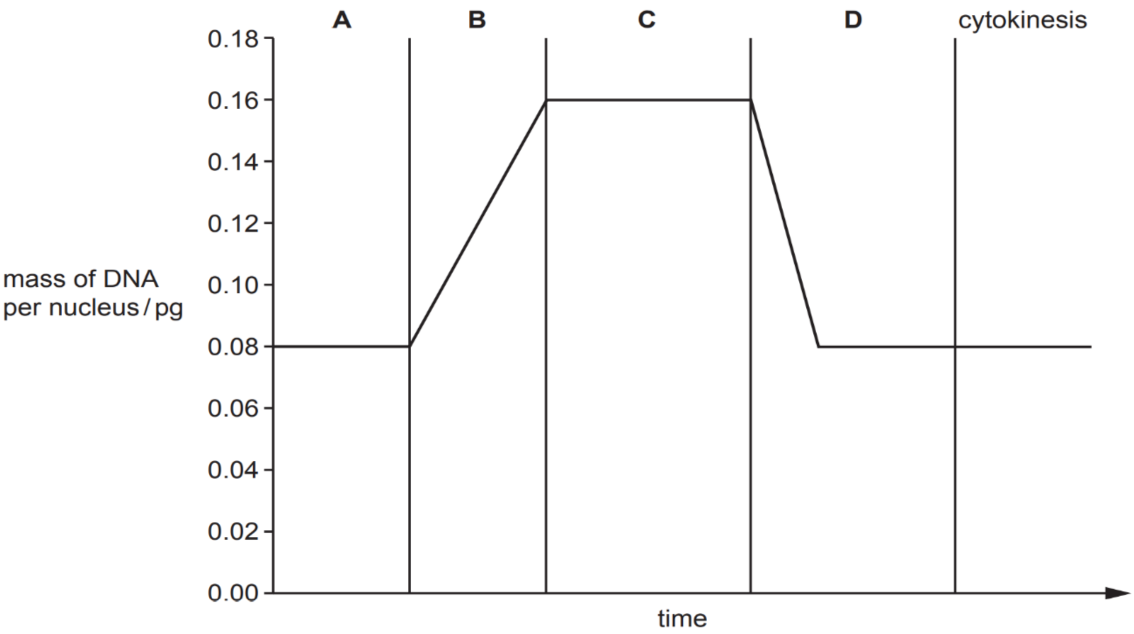


Fig. 4.2

(i) State the phases of interphase shown by **A** and **B** and the stage of mitosis shown by **D** in Fig. 4.2. [1]

1. A = G1 (phase)
2. B = S (phase) [Reject: DNA replication] [Ignore G2 phase]
3. D = Telophase

(ii) Outline what happens in a cell in stage D in preparation for cytokinesis.

…………………………………………………………………………………[2]

Any two:

1. (re)formation of **nuclear envelope** ; **around** each group of **chromosomes** ;
2. **Chromosomes de-condensing** into chromatin [reject: chromatid];
3. (movement of) **organelles** to be **shared** between two daughter cells ;
4. **Spindle**, disassemble / break down / degrades / disappear / AW ;
5. Cleavage furrow forms / cell surface membrane pinches in / constricts / infolds ;

(iii) A stem cell of a female *C. elegans* has 12 chromosomes.

Complete Table 4.1 to show the number of nuclei within the stem cell in stages **A**, **B** and **D** of the cell cycle shown in Figure 4.2 and the number of chromosomes in each nucleus during these stages. [2]

Table 4.1

ANS:

	stage of the cell cycle		
	A	**B**	**D**
number of nuclei within the stem cell	**1**	**1**	**2 ;**
number of chromosomes in each nucleus	**12**	**12**	**12**

1. 1 mark each for correct answer for each row

(iv) Immediately after cytokinesis, daughter cells are **not** identical even though they are genetically identical.

Suggest a reason why daughter cells are not identical immediately after cytokinesis.

…………………………………………………………………………………[1]

1. Unequal sharing out of, cytoplasm / organelles / named organelles, (at cytokinesis) ;
   AVP: cells are of unequal sizes

   [Reject: plasmids / circular DNA in mitochondria, as focus of question is not on genetic factors – refer to qns stem "even though they are genetically identical"]

(b) The tomato plant, *Solanum lycopersicum*, does not tolerate periods of drought (water shortage). Researchers have produced a tomato plant that has an improved tolerance of drought.

The nuclei of plants produce small lengths of RNA known as microRNAs.

MicroRNAs in guard cells have been shown to prevent the synthesis of some proteins.

The guard cells of the drought-tolerant tomato plants produced more microRNA molecules than the guard cells of the non-tolerant plants.

MicroRNA molecules do **not** prevent transcription but interact with messenger RNA (mRNA).

Suggest how this microRNA can interact with mRNA to prevent the production of proteins in guard cells of *S. lycopersicum*.

....[3]

1. microRNA {**binds** to mRNA / forms hydrogen bonds with (bases on) mRNA}
2. bases in microRNA are <u>complementary</u> to bases on mRNA ; [Accept: binds by complementary base pairing]
3. mRNA with microRNA **cannot bind** to (small subunit of) **ribosome** ;
   / anticodon of tRNA cannot bind to (some) codons on mRNA ;
4. AVP ; e.g. complex of microRNA and mRNA recognised for **degrading** ;

[Total: 9]

QUESTION 5

RNA polymerase in *Mycobacterium tuberculosis* is composed of five different polypeptides. Gene *rpoB* codes for one of these polypeptides known as the β-subunit.

One or more mutations in a specific region of *rpoB* result in strains of *M. tuberculosis* that are resistant to rifampicin, an antibiotic. In these strains, mutations often occur in two DNA triplets in this region, in positions 526 and 531.

Table 5.1 summarises the results of an investigation into seven rifampicin-resistant strains, A to G, that have amino acid changes for positions 526 and 531.

Table 5.1 includes:

- the change in the **mRNA codon** for position 526 or position 531
- the amino acid change that has occurred as a result of the mutation
- the minimum concentration of rifampicin required to inhibit growth of the bacterial strain (MIC)
- the number of **other** mutations occurring within the specific region of *rpoB*.

Table 5.1

Key
≈ approximately   ⩾ greater than or equal to   ⩽ less than or equal to

strain	codon involved	mRNA codon change	amino acid change	MIC / $\mu g\,cm^{-3}$	number of other mutations in the specific region
A	526	CAC → UAC	His → Tyr	⩽50	0
B	526	CAC → AAC	His → Asn	⩾100	1
C	526	CAC → CGC	His → Arg	≈50–75	2
D	526	CAC → CGC	His → Arg	⩾100	3
E	526	CAC → CGC	His → Arg	≈50	3
F	526	CAC → UUC	His →	⩾100	3
	531	UCG → UUG	Ser → Leu		
G	526	CAC → UAC	His →	⩾100	3
	531	UCG → UUC	Ser → Phe		

(i) Complete Table 5.1 to show the amino acid changes that have occurred in strains **F** and **G**. [1]

1. F : <u>Phe</u>
2. G : <u>Tyr</u>

(ii) With reference to Table 5.1, list the strains of *M. tuberculosis* that show the greatest resistance to rifampicin.

…………………………………………………………………………………………..[1]

1. **<u>B D F G</u> ;** *in any order*

(iii) Suggest reasons to explain why strains **C and D** show:

- resistance to rifampicin
- different levels of resistances to rifampicin.

…………………………………………………………………………………………[4]

Explanation for resistance in general:

1. **tertiary structure / 3D conformation**, of, **β-subunit** / enzyme's **allosteric** binding **site** (bound by antibiotic) is **altered** ;
2. rifampicin / antibiotic, **cannot / does not, bind** as well ;
   [**REJECT**: if context is binding to active site as inhibition is still possible but at high MIC]

Explanation for different level of resistance: [max 2]

3. (because) *[Data quoting]* **C** has <u>2</u> (other) mutations and **D** has <u>3</u> (other mutations that are involved) ;
4. Mutation may result in **different extent of changes** to, **structure** / 3D conformation of β-subunit / enzyme, and result in different, effects / levels of resistance ;
5. [different locations of the mutations] Ref. C has other mutations which may be found **outside** the allosteric site *(this site is not at the active site)* ; **D** has 3 (other) mutations which may be found, **in** the allosteric site of the enzyme *(resulting in D's higher resistance to antibiotics)*
6. [effects of different amino acid changes in allosteric site] Ref. **Neutral mutations** in <u>C</u> (similar chemical property of amino acid) ; **missense mutations** in <u>D</u> (different chemical property of amino acid) ;
   [reject: nonsense mutations]

[Total: 6]

QUESTION 6

Skin colour in humans is classified according to the Von Luschan's chromatic (colour) scale as seen in Table 6.1.

Table 6.1

Colour	Sunburning	Tanning behavior	Von Luschan's chromatic scale
Light, pale white	Always	Never	0–6
White, fair	Usually	Minimally	7–13
Medium white to light brown	Sometimes	Uniformly	14–20
Olive, moderate brown	Rarely	Easily	21–27
Brown, dark brown	Very rarely	Very easily	28–34
Very dark brown to black	Never	Never	35–36

(a) With reference to Table 6.1, state and explain the type of variation depicted by the trait of skin colour.
………………………………………………………………………………………..…………[2]

Type of variation: ……………………………..

1. Continuous variation

Explanation: ………………………………………………………………………………

1. Ref. **range** in the chromatic scale in each category / type; **+ any quote** of range for any colour eg. in light, and pale white, the scale ranges from 0-6.
2. Effect of environmental factors e.g. amount of sunlight exposure on the phenotype;
   [Any one]

(b) Fur colour in cats is also controlled by genes.

The gene coding for colour is located in the X chromosome. Allele O gives rise to orange fur, while allele B gives rise to black fur. Both allele O and allele B are co-dominant to each other. A cat with both alleles have a tortoise shell variegation.

Another gene E on an autosome (non-sex chromosome) affects the overall colour of the fur. One dominant allele of this gene (allele E) prevents normal development of melanocytes (pigment producing cells). Therefore, cats with genotype (E_) will have entirely white fur regardless of the genotype at the gene locus for colour.

(i) State the type of gene interaction between the two genes.
..………………………………………………………………………………………..…………[1]

Type of interaction: ……………………………..

1. (dominant) epistasis

(ii) A female cat with tortoise shell variegation [X^OX^Bee] and a white male cat [X^BYEe] which is the offspring of a black female cat [X^BX^Bee] are chosen to mate.

Draw a genetic diagram to show the outcomes of this cross.

…………………………………………………………………………………………………[4]

Key
X^O – X chromosome carrying the allele coding for orange colour
X^B – X chromosome carrying the allele coding for black colour
Allele O and allele B are co-dominant.
Y – Y chromosome (does not carry allele for colour)

Allele E code for protein that prevents normal development of melanocytes
Allele e code for non-functional protein that does not prevent normal development of melanocytes.
Allele E is dominant to allele e.

Parental phenotype	Tortoise shell female	x	White male
Parental genotype	X^OX^Bee	x	X^BYEe
Parental gametes	X^o e, X^B e	x	X^BE, X^Be, YE, Ye

Punnett square

	X^o e	X^B e
X^BE	$X^o X^BEe$ White female	$X^B X^BEe$ White female
X^Be	$X^o X^Bee$ Tortoise shell female	$X^B X^Bee$ Black female
YE	X^o Y Ee White male	X^B YEe White male
Ye	X^o Yee Orange male	X^B Yee Black male

F_1 phenotypic ratio: 2 white female : 1 tortoise shell female : 1 black female : 2 white male : 1 orange male: 1 black male

Mark scheme:
1. Parental genotypes ;
2. Parental gametes (**circled**) ;
3. F1 genotypes / Punnett square ;
4. F1 genotypes correspond to phenotypes ;
5. F1 phenotypic ratio

[Max 4]

Marker's guidance: no marks if allele is not on X chromosome; max 3m for small symbols errors e.g. superscripting E/e on a capital letter.

(c) A chi-square analysis was performed on another group of cats. There was a total of six phenotypic classes in the F1 generation.

chi-squared table:

Deg. of freedom	**p= 0.5**	**p= 0.1**	**p= 0.05**	**p= 0.01**	**p= 0.001**
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.36	7.78	9.49	13.28	18.46
5	4.35	9.24	11.07	15.09	20.52

The calculated chi-square value for the test is found to be 8.15.

(i) Use this information and make some valid conclusions for this analysis.
……………………………………………………………………………………………[3]

1. [Comparing χ^2] Since the calculated χ^2 value 8.15 is **smaller** than **critical χ^2** value 11.07 at p = 0.05; (at DoF= 6-1 = 5)

   OR [Comparing p values] since the P value corresponding to χ^2 value 8.15 is between 0.1 and 0.5, which is larger than benchmark of p=0.05;

2. there is no significant difference between the observed and the expected values at the 5% level in each phenotypic class. Any difference is due to chance;

3. The observed numbers fit the expected phenotypic ratio relating to epistasis;

(ii) Suggest a way to ensure that the data obtained from observing the results of the cross is statistically reliable.
……………………………………………………………………………………………[1]

1. Observe a large number of progeny
2. AVP e.g. more repeat experiments to obtain more sets of data

[Total: 11]

QUESTION 7

Each human cell contains approximately 2 meters of DNA if stretched end-to-end; yet the nucleus of a human cell, which contains the DNA, is only about 6 μm in diameter.

(a) Describe how DNA is packaged to fit into the nucleus of eukaryotic cells.
....[4]

1 Nucleosomes formed by the winding of **negatively charged chromosomal DNA** around **basic/positively charged histones;**

2 Nucleosomes and linker DNA result in the formation of a 10nm **chromatin fibre** / **"beads-on-a-string"** structure.

3 10nm chromatin fibre **coils around itself** to form a 30 nm chromatin fibre / solenoid

4 6 nucleosomes in **one turn of the helix** of the solenoid

5 Attachment of the 30 nm chromatin fibre to a {**central protein scaffold / nuclear matrix}** to form radial loops / 300 nm chromatin fibre;

6 **Further packing** / condensation of the 300nm chromatin results in the formation of a metaphase chromosome

(b) The ends of each DNA molecule are lengths of DNA called telomeres.

In humans, telomeres are made up of the base sequence TTAGGG, which is repeated up to 400 times.

In some cells, each time a DNA molecule is replicated, the last 150 - 200 bases are not copied. The telomeres therefore get shorter with each DNA replication until they become too short for DNA replication to take place.

Calculate the maximum number of replications that can occur with the DNA that contains 400 TTAGGG repeats.

Show your working.

....[2]

1 **Total number of bases** in 400 repeats = 400x 6 = 2400 ;

2 Maximum number of replications = 2400 / 150 = 16 ;

Eukaryotic DNA contains both introns and exons.

Figure 7.1 shows the structure of a length of pre-mRNA formed from one gene.

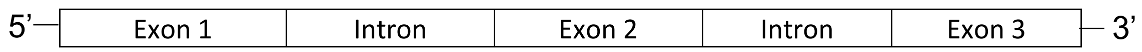


Fig. 7.1

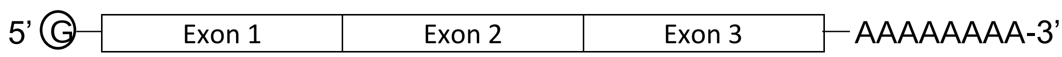
(c) (i) Explain how the pre-mRNA molecule shown in **Fig. 7.1** is converted into a mature mRNA molecule.

....[3]

1. Addition of a {**modified (methylated)** guanine nucleotide / methylated GTP}; to the 5' end of the pre-mRNA ; with the help of guanyltransferase**.**
2. Recognition and binding of spliceosomes to splice sites; **introns are excised ; exons are ligated during** (RNA) **splicing;**
3. Addition of a {**long chain** of adenine nucleotides / poly(A) tail} to 3' end of the pre-mRNA transcript ; with the help of poly(A) polymerase;

(ii) Sketch the longest mature mRNA molecule that can be formed from the pre-mRNA shown in Fig. 7.1.

....[1]



1 5' G cap; 3 exons; 3' poly(A) tail

The processing pre-mRNA molecule to a mature mRNA molecule does not occur in prokaryotes.

(iii) Explain two other differences in the control of gene expression between eukaryotes and prokaryotes.
....[2]
[Any 2]

	Bacterial Control	**Eukaryotic Control**
1) DNA / Chromatin Level	DNA and **histone modifications cannot occur** ; as prokaryotic DNA are not organized into chromatin / **not associated with histones**	DNA and **histone modifications can occur** ; as eukaryotic DNA is **complexed with histones** and other proteins to form chromatin
2)Transcriptional Level	**Simple** interactions between transcriptional regulatory **proteins** and **DNA** sequences upstream of the cluster of structural genes to regulate **initiation** of transcription	**Complex** interactions between **protein** factors and **DNA** sequences upstream of gene (i.e. control elements) to initiate and regulate **rate** of transcription
2)Transcriptional Level	Functionally related genes **are transcribed together as operons** / all genes of operon controlled by only one promoter	**No operon** / each gene has own promoter
~~**3) Post - Transcriptional Level**~~	Reject: due to Qn requiring *other* differences ~~**Primary transcripts are the actual mRNA / no post-transcriptional modification done;**~~ ~~**Lower stability** of transcript~~ ~~/ degradation within seconds or minutes~~ ~~/ mRNAs have shorter half-life to rapidly respond to environmental changes~~	Reject: due to Qn requiring *other* differences ~~Presence of 5' capping / 3' Poly(A) tailing;~~ ~~**Higher stability** of transcript due to~~ ~~/ prevent fast degradation~~ ~~/ mRNAs have longer half-life, remaining much longer to orchestrate protein synthesis prior to their degradation by nucleases in the cell~~
4)Translational Level	Control at translational level is **unlikely** ; due to **simultaneous transcription and translation**	**Any 1 example:** **Phosphorylation** of (ribosomal) <u>translation initiation factors</u> can inhibit or increase translation **Regulatory proteins** bind to <u>5' UTR</u> ; **preventing ribosomal attachment** and decreasing translation rate

		RNA interference / miRNA / siRNA prevents translation or degrades mRNA
5) Post - Translational Level	**No / minimal** post-translational modifications ; due to lack of ER / Golgi apparatus (membranous organelles)	Presence of ER / Golgi apparatus (membranous organelles); Newly synthesized proteins need to be **modified** to form functional proteins; + any 1 example below ~ Proteolysis ~ Biochemical modifications ~ Selective protein degradation via ubiquitination ~ Transport of proteins to target destinations in the cell where it functions

[Total: 12]

QUESTION 8

(a) Coronavirus disease 2019 (COVID-19) is a pandemic affecting many countries globally. COVID-19 is caused by a novel coronavirus, severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Figure 8.1 shows the structure of SARS-CoV-2.

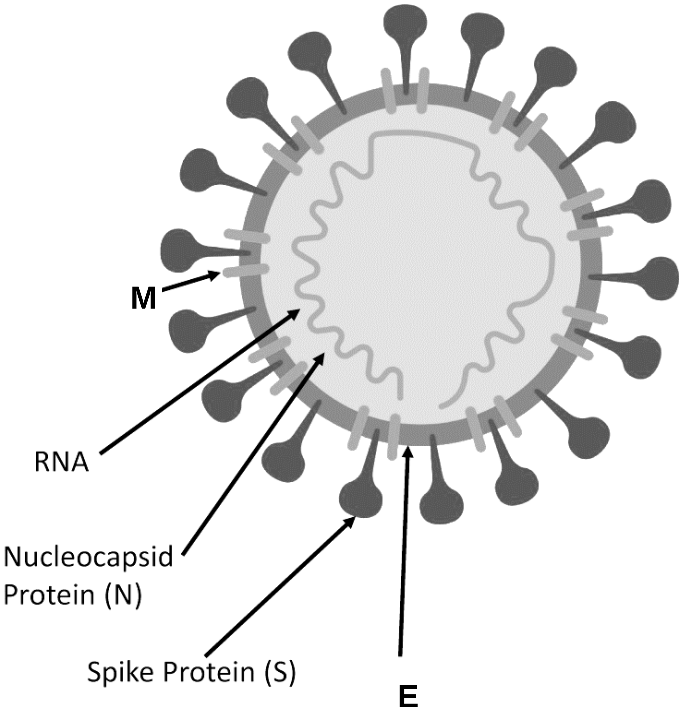


Fig. 8.1

(i) Identify the monomer(s) of structure **M** and the main constituent of structure **E**.

monomer(s) of structure **M**: ……………………………………..

main constituent of structure **E**: ……………………………………..

[2]

Ans:

monomer(s) of structure **M**: Amino acids ; Accept: monosaccharides; [Reject: glucose as this is too specific]

main constituent of structure **E**: Phospholipids [Reject: proteins as they are not the main constituent; Reject "lipids" as this is too vague] ;

Figure 8.2 shows a simplified version of the reproductive cycle of SARS-CoV-2.

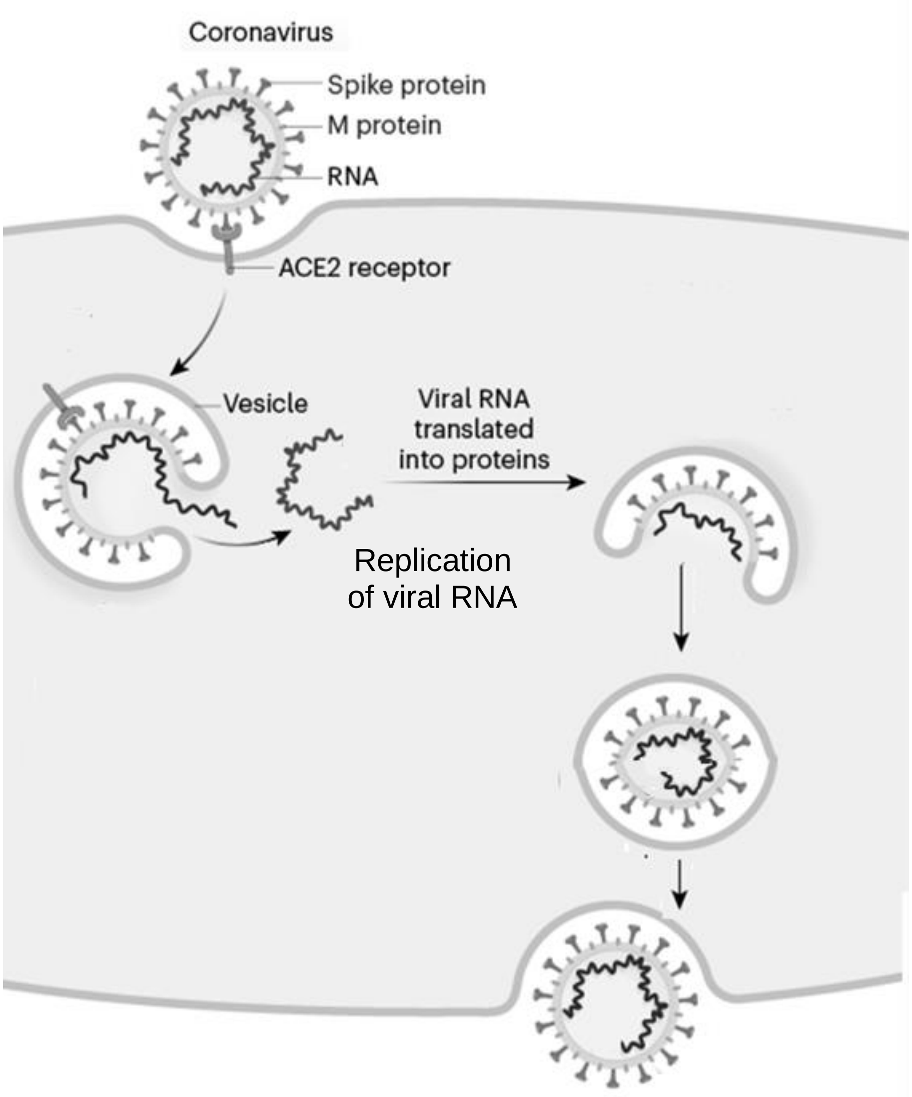


Fig. 8.2

(ii) With reference to Fig 8.2 and your biological knowledge, compare the reproductive cycle of SARS-CoV-2 with that of influenza virus.
………………………………………..……………………………….[3]

1 similarity + 2 differences

<table>
<tr><th>Feature</th><th>SARS-CoV-2</th><th>Influenza virus</th></tr>
<tr><td>1 Entry

[Similarity]</td><td colspan="2">Both viruses enter via <u>receptor mediated endocytosis</u>
/
For both viruses, viral <u>envelope</u> fuse with endosome <u>membrane</u> to release the Nucleocapsid/viral RNA</td></tr>
<tr><td>2 Adsorption

[Difference]</td><td><u>Spike protein</u> binds to <u>ACE2</u> on host cell</td><td><u>Haemagglutinin</u> binds to <u>sialic acid</u> (receptor) on host cell</td></tr>
<tr><td>3 Glycoprotein embedding

[Difference]</td><td>Comparing the time of insertion of viral glycoproteins into membrane
/
Glycoproteins are not embedded into cell surface membrane
/ synthesized and embedded into ER's / Golgi body's membrane</td><td>Comparing the time of insertion of viral glycoproteins into membrane

/
Glycoproteins embedded into cell surface membrane</td></tr>
<tr><td>4 Release of new viruses

[Difference]</td><td>New viruses are released via <u>exocytosis</u>,</td><td>New viruses are released via <u>budding</u></td></tr>
<tr><td>5 AVP</td><td></td><td></td></tr>
</table>

(b) It was discovered that the SARS-CoV-2 genome was positive sense RNA and that its capsid also contained viral RNA dependent RNA polymerase.

Based **solely** on the information provided in this question, suggest how new variants of SARS-CoV-2 are formed.
…………………………………………..……………………………..……….[3]

1. By <u>antigenic drift</u> [Reject: genetic drift]
2. spontaneous **mutations** occur in RNA **genome** [ignore: genes, which usually means DNA] coding for viral antigens e.g. spike protein ; due to **lack of proof-reading in viral RNA-dependent RNA polymerase** ;
3. thus resulting in (minor) **change in 3D conformation** of **spike protein,** a new virus strain results.

(c) COVID-19 is commonly detected in individuals via swab tests, which involve the use of real time Reverse Transcription Polymerase Chain Reaction (RT-PCR). A sample is first collected from parts of the body where SARS-CoV-2 gathers, such as a person's nose or throat. The sample is then broken down to expose the genetic materials and RT-PCR is then performed to determine the viral load.

(i) Explain why both reverse transcription and PCR are needed in order to detect the virus.

……………………………………………………..……..……………………………..[2]

1. Reverse transcription to **synthesis the viral** <u>DNA</u> from the viral <u>RNA</u> **genome** first ; [Reject: convert]
2. PCR is then needed to **amplify** the small amount of viral DNA (after RT) for analysis purposes ;

(ii) Four individuals, W X Y & Z were placed under quarantine order by the Ministry of health. After 13 days, they took the exit swab test (involving RT-PCR). Subsequently, gel electrophoresis was carried out on the respective RT-PCR products synthesised and the result is shown in Figure 8.3.

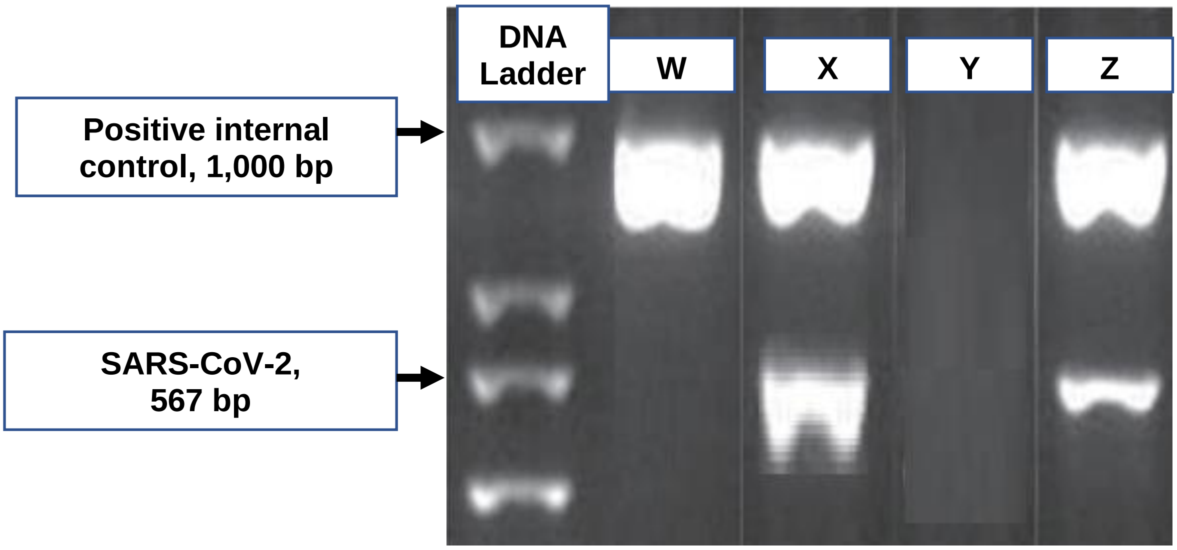


Fig. 8.3

Based on Fig. 8.3, identify which individual/s could be released from quarantine. Explain your answer.

………..……..……………………………………………………………………....[2]

1. (Only) individual <u>W ;</u>
   [**Reject** Y - Y's sample failed to display result for positive control, highlighting that the result is not reliable for Y]

2. Presence of band for positive control / 1000bp band; absence of band for SARS-CoV-2 / 567bp band;

[Total: 12]

QUESTION 9

Figure 9.1 shows the arrangement of bones in the pentadactyl forelimbs of four vertebrates. This is used by many scientists to provide evidence for evolution.

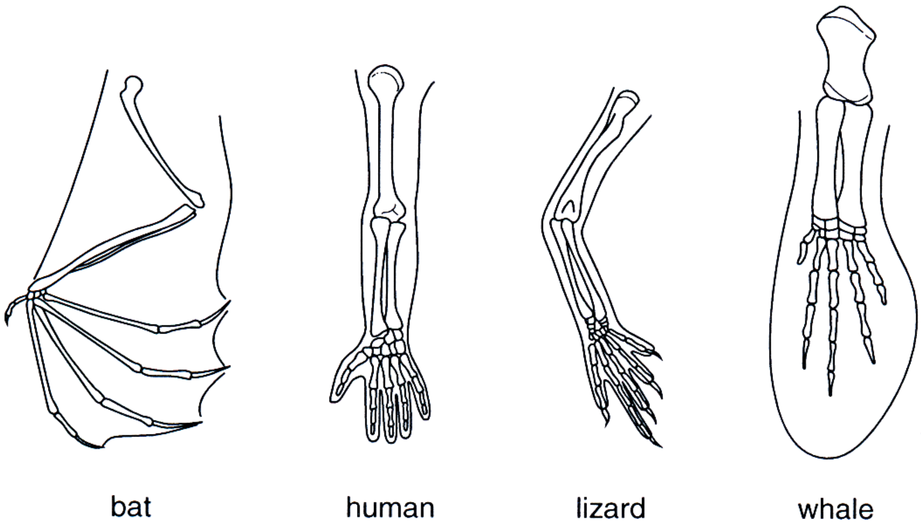


Fig 9.1

(a) (i) State the term used to describe the relationship between structures such as those in Fig. 9.1.

....[1]

1 **Homologous** structures

(ii) Explain how the relationship between the structures in Fig. 9.1 provides evidence to support the theory of evolution.

....[3]

[Darwin's theory of evolution focuses on descent with modification, where structures are inherited from a common ancestor, but modified to adapt to different environments/ natural selection pressures.]

1 Same **arrangement of bones** in all forelimbs shown / all have pentadactyl forelimbs;

2 Forelimbs serve **different functions** ; [Reject: complete loss of function / vestigial structure]

3 Reveals **descent with modification** / structure is likely to have originated from a **common ancestor** and **modified** over time;

4 {from **natural selection** / to adapt to different environments/ natural selection pressures};

(b) Suggest how natural selection could have brought about the evolution of the skeleton of a bat's wing.
....[3]

1 **(Genetic and phenotypic) Variation** in the skeleton of the bat's wings ; resulted from spontaneous (DNA) <u>mutations</u> ;

2 Ref. Selection pressures; + E.g.;

3 Bats with **this skeleton structure** have a **selective advantage** ;

4 Allowing them to **survive till reproductive age and reproduce** ; passing <u>alleles</u> [Reject: pass down traits due to Lamarck's theory] to offspring ; **increasing allele frequency over time;** [to address the term "evolution" in the question stem]

When comparisons were made with *cytochrome c* gene of the four vertebrates, there was strong evidence to show that these vertebrates share a common ancestry.

(c) (i) Explain how molecular homology could be used to support the claim that these vertebrates share an evolutionary relationship.
....[2]

1 [Method] Direct **comparison of DNA sequence** of a **common gene** (cytochrome c gene). [Reject: amino acid sequence of cytochrome c gene / common DNA sequence which is non-coding due to context given] [Reject: preserved genes]

[How to analyse]

2 Organisms that show **high similarity** in their sequences indicate that they probably **share a common ancestor.**

/ The more similar the sequences of the organisms are, the more closely related they are in their evolutionary relationship;

3 Number of mutations in genetic sequence is used to calculate the length of time since divergence

Scientists also make use of fossil records to see the evolution of species through the modification of homologous structures from a common ancestor to the present descendant through a series of transitional forms.

However, the use of these fossils to support Darwin's theory of evolution may be inconclusive.

(ii) Explain why this is so and how molecular data is able to overcome the limitations of the study of fossil records.
....[2]

[Advantage of molecular data - limitation of fossil records]

1 **Molecular** data such as nucleotide and amino acid sequences are **quantifiable**, in **abundance** and **open to statistical analysis** ; while **structural** differences in the fossils are **limited** in quantity.

2 Molecular data are **unambiguous** / **objective**; while analysing anatomical differences between the fossils may be **subjective**.

3 (Molecular data provides a clear model of evolution by comparing the nucleotide and amino acid sequence because the rate of molecular change in genes and proteins is regular like a molecular clock);

Ref. not affected by convergent evolution (as not looking at morphology); whereas similarities in fossil structures may be **analogous** structures / due to <u>convergent</u> evolution (in distantly related organisms / organisms not sharing a common ancestor recently, thus resulting in inaccurate classification)

4 AVP

[Total: 11]

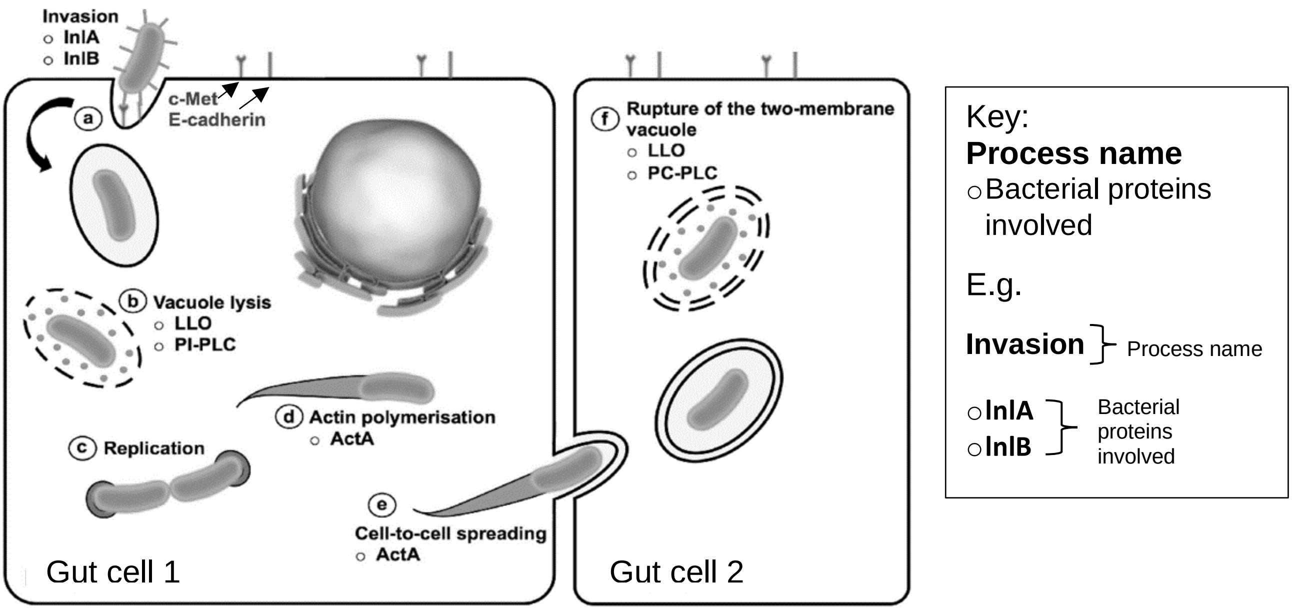
QUESTION 10

Listeria monocytogenes is a type of bacteria that is commonly found in water, soil, and faeces. Humans are infected when they consume foods contaminated with the bacteria.

The bacterium first invades the human host through entering the epithelial cells lining the gut.

It has the capacity to then cross the intestinal barrier and also the blood–brain barrier and the placental barrier, infecting the brain and placenta as well to cause encephalitis (inflammation of the brain), and spontaneous abortions in pregnant women.

Figure 10.1. shows the reproductive cycle of the bacterium in the host cell. This bacterium is known to form a "comet tail" using host cell actin to spread to other host cells (Bonazzi *et. al,* 2009).



Antimicrobial Resistance in Listeria Species - Scientific Figure on Research Gate. Available from: https://www.researchgate.net/figure/L-monocytogenes-intracellular-life-cycle-a-Listeria-invades-the-host-cells-via-a_fig2_326528987

Fig. 10.1

(a) Suggest why *L. monocytogenes* can also specifically cause disease symptoms in the brain and induce spontaneous abortions in its human host.

……………………………………………………………………………………………[1]

1. Presence of {receptor proteins / c-MET and E-cadherin}, on the brain and placenta cells;
   which have 3D conformation complementary to that of {bacterial adhesion proteins / InlA and InlB};

(b) With reference to Fig. 10.1, explain one way in which the invasion by *L. monocytogenes* can affect the gut cells.

…………………………………………………..……………………………………..….[1]

1. **Ref. interfering with gut cell functions; e.g.** {processes relevant to actin such as movement of vesicles, cell support etc / competes for nutrients e.g. nucleotides, amino acids etc with the gut cell for its own binary fission / protein synthesis}
   [Reject: hijacks host enzymes due to bacteria being alive with its own enzymes / host eukaryotic ribosomes due to host using 80s ribosomes]
   / AVP e.g. use the host ATP to propel the actin tails; **less ATP to provide energy for host metabolic processes**

2. AVP

(c) With reference to your knowledge of the adaptive immune system, outline how the bacterial-infected gut cells may work together with lymphocytes to respond to the infection.

……………..……………………………………………………………………...………[2]

1. Infected gut cells can **present antigen** on <u>MHC</u> class I;
2. <u>Cytotoxic T cells</u> recognize and bind to the bacterial antigens on host cells; leads to **cytolysis** [Reject: plasmolysis; hydrolysis] of infected T cells
3. via secretion of <u>perforin</u>, (which form pores in the infected cell membranes and disrupt membrane integrity) and <u>granzymes</u> (which enter the infected cells to break down protein and trigger cell death by apoptosis);
4. Ref. <u>T helper cells</u> activating CD8 T cells by secreting <u>cytokines</u>;

[Reject: humoral response route]

(d) Assuming that a *L. monocytogenes* cell has a spontaneous mutation in one of its chromosomal genes to evade the immune system, list two processes that can facilitate the conferring of this evasion between bacteria cells.

………………………………..……………………………………………………..…………[1]

Process 1 ……………………………..   Process 2 ……………………………..

1. Binary fission / transformation / transduction; [Any 2]
   [Accept: conjugation as HFr mode does involve chromosomal partial transfer but this is out of syllabus]

(e) Antibiotics such as penicillin are generally prescribed for bacterial infections by *L. monocytogenes.* Explain why penicillin is effective.

…………………………………………………..…………………………………..…………[2]

1. Ref. *L. monocytogenes* having a <u>peptidoglycan</u> cell wall; [Reject: peptidyl transferase]
2. Penicillin is a (irreversible) **competitive inhibitor** of bacterial enzyme <u>transpeptidase</u>;
3. **Cross-links** between peptidoglycans **do not form**, leading to **weakened cell walls**, the increased turgor pressure when **water enters** bacteria cell by osmosis,

causes cell to **burst**. [Reject: cell wall is broken down, as this inhibition only prevent new cross linkages from forming]

[Total: 7]

End of Paper